Nose Wheel Steering Actuator – General Data

1. General

The nose wheel steering actuator consists of a steering valve, cylinder, piston and rod assembly. The actuator is hydraulically operated. Movement of the steering control lever in the flight compartment rotates the drum and changes the position of the steering valve. This action directs hydraulic pressure to one side of the piston and rod, moving the piston outboard and rotating the steering linkage and torque links which turn the nose wheel. Internal stops in the steering actuator limit the arc of travel. The actuator also acts as a shimmy damper during taxiing, take-off, and landing runs. A 10 micron in-line filter at the pressure inlet protects the internal components of the actuator against damage from contaminated hydraulic fluid.

2. Servicing

A. Service Hydraulic Filter

1) Discharge damping accumulator pressure by operating wing flaps.

2) Place container to receive spillage and remove pressure tube assembly.

3) Cut lockwire and remove filter, using Heroux special tool T–189.

4) Disassemble filter and examine filter element for damage.

5) Clean filter element by back-flushing with cleaning solvent and drying with clean compressed air.

Note Ultrasonic cleaning equipment may be used, if available, to clean filter element.

6) Examine O-rings for serviceability and replace as necessary. O-ring on filter element is P/N MS28775–010, and O-ring between end cap and housing is P/N MS28778–6.

7) Assemble filter element in housing. Install and tighten end cap.

8) Inspect packing for serviceability and replace as necessary with new packing P/N MS28778–4. Install packing and filter to actuator using special tool T–189. Ensure that direction of flow arrow on filter body points towards actuator.

9) Lockwire filter to actuator body using MS20995C20.

10) Install pressure tube assembly.

11) Bleed and function test nose wheel steering system. (Refer to 32-50-00, Nose Wheel Steering System – General Data).

3. Removal/Installation

Refer to Figure 1 for details.

A. Remove Nose Wheel Steering Actuator

Note Removal can be carried out with aircraft on the ground or in the jacked-up position.

1) Prepare aircraft as follows:

a) Aircraft on ground:

Position aircraft nose wheel on greased plates or similar arrangement. Ensure parking brake is on, or main wheels are securely chocked.

Note Two smooth sheets of steel plate with grease applied between them is a satisfactory platform for nose wheel actuator removal and tests. The upper plate should turn smoothly with wheel when steering is operated without any movement between the nose wheel tire and the upper surface of the plate.

b) Aircraft on jacks:

Remove quick-release pin connecting upper and lower torque links on shock strut. Move upper torque link as required, push center latch pin against spring so that latch pin is flush with lower surface and protrudes the upper surface; secure latch pin at upper surface using a hose clamp (jubilee clip). Install quick-release pin at upper and lower torque links.

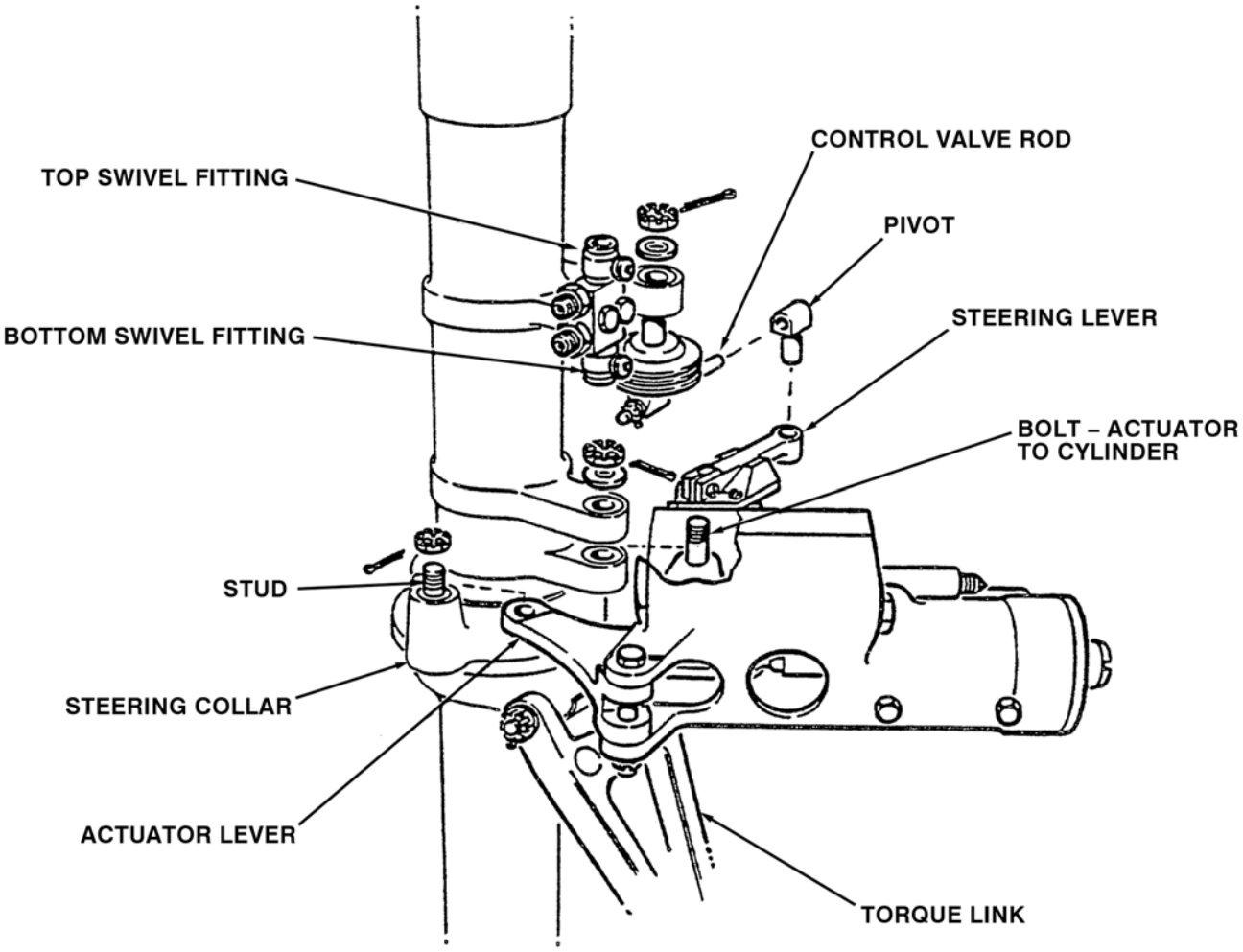
2) Discharge hydraulic system accumulator by operating wing flaps.

3) Open nose baggage compartment. Remove screens and covers as required.

4) Remove cover on underside of fuselage at nose gear leg.

5) Disconnect hydraulic return line at top swivel fitting and hydraulic pressure line at bottom swivel fitting on nose gear leg. Cap openings and lines.

6) Remove cotter pin, nut and washer attaching actuator lever to steering collar. Support actuator and remove cotter pin, nut and washer from bolt attaching actuator to cylinder. Turn leg to obtain clearance from steering collar and torque links and remove bolt. Turn leg fully to right, disengage pivot on steering lever from control valve rod, and lift actuator clear of stud on steering collar.



Nose Wheel Steering Actuator Installation
Figure 1

B. Install Nose Wheel Steering Actuator

1) Position and support actuator at nose gear leg. Turn leg fully to right, locate actuator lever on stud of steering collar, and engage pivot on steering lever with control valve rod. Align actuator with cylinder and insert bolt. Secure bolt with washer, nut and cotter pin.

2) Install washer, nut and cotter pin to stud on steering collar.

3) Connect hydraulic pressure line to bottom swivel fitting and hydraulic return line to top swivel fitting on nose gear leg.

4) Check actuator range of movement and adjust if necessary. (Refer to Para 4.B).

CAUTION **BEFORE CHECKING THE ACTUATOR FOR RANGE OF MOVEMENT, ENSURE CORRECT ADJUSTMENT OF THE STEERING ACTUATOR. INCORRECT ADJUSTMENT CAN CAUSE THE FRACTURE OF THE STEERING COLLAR DUE TO TOO MUCH RANGE OF MOVEMENT OF THE ACTUATOR LEVER (REFER TO Para 4.B).**

5) Bleed, function test, and check rigging of nose wheel steering system. (Refer to 32-50-00, Nose Wheel Steering System – General Data).

6) If installation was carried out with aircraft on jacks, remove hose clamp (jubilee clip) from latch pin at upper torque link. Lower aircraft to ground.

7) If installation was carried out with aircraft on ground, remove greased plates.

8) Install cover on underside of fuselage at nose gear leg.

9) Install screens and covers as necessary. Close nose baggage compartment.

4. Adjustment/Test

A. Bench Pressure Test

Refer to Figure 2 for details.

Equipment required:

- Hydraulic test rig, two pressure supplies 2250 psi, and flow rate 1.6 gpm (U.S.)
- Fluid MIL–H–5606
- Flowmeter to check flow rate of 1.2 to 1.6 gpm (U. S.)
- Torque wrench to 100 pound–inches

1) Filling and bleeding:

a) Remove two test port plugs, and connect test ports A and B together with shorting line.

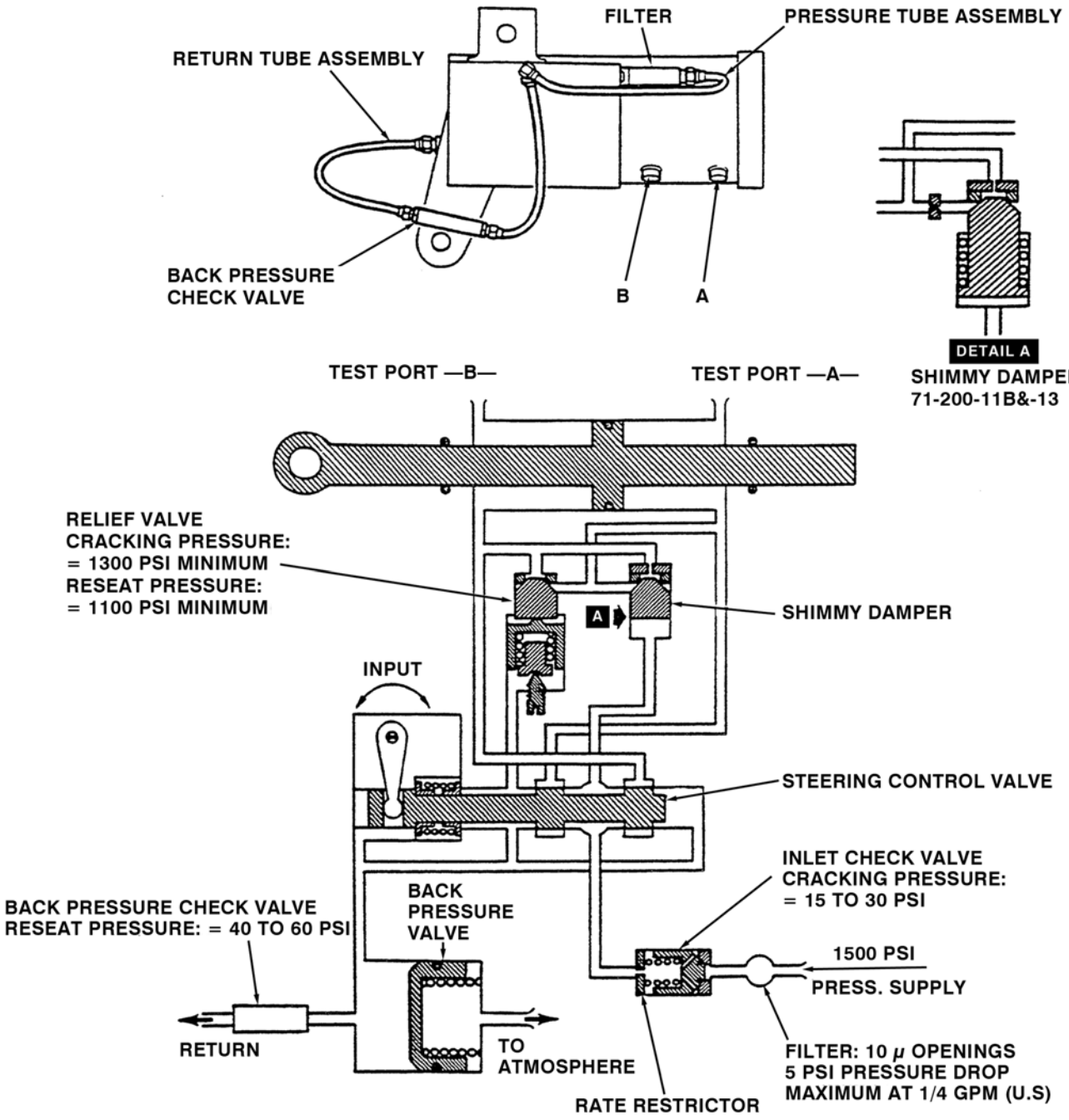
b) Connect hydraulic test rig supply line to pressure tube assembly and return line to the return tube assembly.

c) Place actuator so that input swivel assembly is facing down.

d) Apply pressure of 1500 psi, turn lever of input swivel assembly fully in one direction and hold for 10 seconds minimum. Turn lever fully in opposite direction and hold for 10 second minimum.

e) Repeat Para 4.A.1).d) a minimum of five times.

f) Place actuator so that input swivel assembly is facing up, then repeat Para 4.A.1).d), and Para 4.A.1).e).



Test Ports and Schematic Hydraulic Diagram
Figure 2

2) Cracking pressure, inlet check valve:

   a) Connect hydraulic test rig supply line to pressure tube assembly. Remove test port plugs and leave test ports open.

   b) Turn lever of input swivel assembly fully in either direction and gradually increase supply pressure until a small steady flow is obtained at test port. Check that supply pressure is 15 to 30 psi.

3) Reseat pressure, back pressure check valve:

   a) Open pressure tube assembly to atmosphere. Connect hydraulic test rig supply line to test port A.

   b) Place actuator so that input swivel assembly is facing up.

   c) Turn lever of input swivel assembly fully counterclockwise (looking from above). Raise supply pressure until a steady flow is obtained at return tube assembly. Decrease supply pressure gradually until flow stops. Check supply pressure is 40 to 60 psi at reseat.

   d) Maintain reseat pressure. After 3 minutes, external leakage must not exceed five drops per minute at each port.

4) Relief valve and shimmy damper:

   a) Place lever of input swivel assembly in neutral position (control valve centered). Connect hydraulic test rig supply line to pressure tube assembly and apply pressure of 1500 psi.

   b) Connect test port A to a regulated supply line. Raise regulated pressure until flow is increased over leakage flow at test port B. Cracking pressure must not be less than 1300 psi.

   c) Lower regulated pressure at test port A gradually until flow at test port B is reduced to leakage flow. Reseat pressure must not be less than 1100 psi.

   d) Connect flowmeter at test port B. Increase regulated pressure at test port A until 1.2 gpm (U.S.) is obtained at test port B. Check that pressure does not exceed 1800 psi.

   e) Connect regulated supply line to test port B and flowmeter to test port A. Repeat tests similar to Para 4.A.4).a) through Para 4.A.4).d).

   f) Release pressure supply at pressure tube assembly and disconnect supply line. Leave pressure tube assembly open to atmosphere to permit bypass through shimmy damper orifice.

   g) Apply regulated pressure of 1000 psi at test port B. Flow at test port A must be 1.4 to 1.6 gpm (U.S.).

5) Control valve:

   a) Connect test ports A and B together with shorting line. Connect hydraulic test rig supply line to pressure tube assembly. Place lever of input swivel assembly in neutral (control valve centered).

   b) Apply pressure of 1500 psi. Leakage flow at return tube assembly must be 12 to 20 cc per minute.

   c) Torque required at lever of input swivel assembly to crack control valve must not exceed 65 pound–inches, and the flow must stop on reduction to 40 pound–inches.

   d) Disconnect test ports A and B and install packing and plug at each port. Connect hydraulic test rig return line to return tube assembly.

   e) With supply pressure at 1500 psi, apply a torque of 75 pound–inches; extension or retraction rate must be 7 ± 1 seconds for a complete stroke.

f) Position piston at center of stroke. With control valve centered, piston creep rate must be less than 0.030 inch per minute.

6) Proof pressure:

a) With plugs installed at test ports A and B and return tube assembly capped, apply a pressure of 2250 psi at pressure tube assembly at a rate of not more than 25,000 psi per minute (400 psi per second).

b) Turn lever of input swivel assembly slowly clockwise and hold for two minutes.

c) Turn lever of input swivel assembly slowly counterclockwise and hold for two minutes.

d) There must be no leakage or permanent set.

Note A slight wetting insufficient to form a drop is acceptable.

e) Release pressure completely and then apply 40 psi at pressure tube assembly. There must be no leakage.

f) Release pressure and disconnect actuator from test rig. Cap pressure and return tube assemblies unless actuator is to be installed immediately.

B. Adjust Actuator Range of Movement

Equipment required:

- Heroux special tool T–186
- Heroux special tool T–194

1) Adjustment of actuator range of movement can be accomplished with the aircraft on the ground or in the jacked-up position; refer to Para 3.A.1) for preparation method.

2) Open nose baggage compartment. Remove screens and covers as required.

3) Remove lockwire from lock key of jam nut on piston rod end. Disengage lock key and loosen jam nut using special tool T–194.

4) Turn leg assembly fully to the right. Turn piston to adjust rod end so that with piston firmly against internal stop, a minimum clearance of 0.010–inch and a maximum clearance of 0.020–inch is obtained between the actuator lever and the outer cylinder of the shock strut. Tighten jam nut on piston rod end using special tool T–194. Engage lock key, and lockwire, using MS20995C20.

5) Turn leg assembly fully to the left. Check that with piston firmly against internal stop (end cap on manifold), a minimum clearance of 0.010–inch and a maximum clearance of 0.020–inch is obtained between stud lug on steering collar and lugs on outer cylinder. If necessary, cut lockwire, disengage lock key and nut from end cap, and adjust end cap using special tool T–186 to obtain clearance. Finger tighten nut on end cap, engage lock key and lockwire using MS20995C20.

6) If adjustment was carried out with aircraft on jacks, remove hose clamp (jubilee clip) from latch pin at upper torque link. Lower aircraft to ground.

7) If adjustment was carried out with aircraft on ground, remove greased plates.

8) Install screens and covers as necessary. Close nose baggage compartment.

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